

Introduction to Python

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What is a Programming Language?

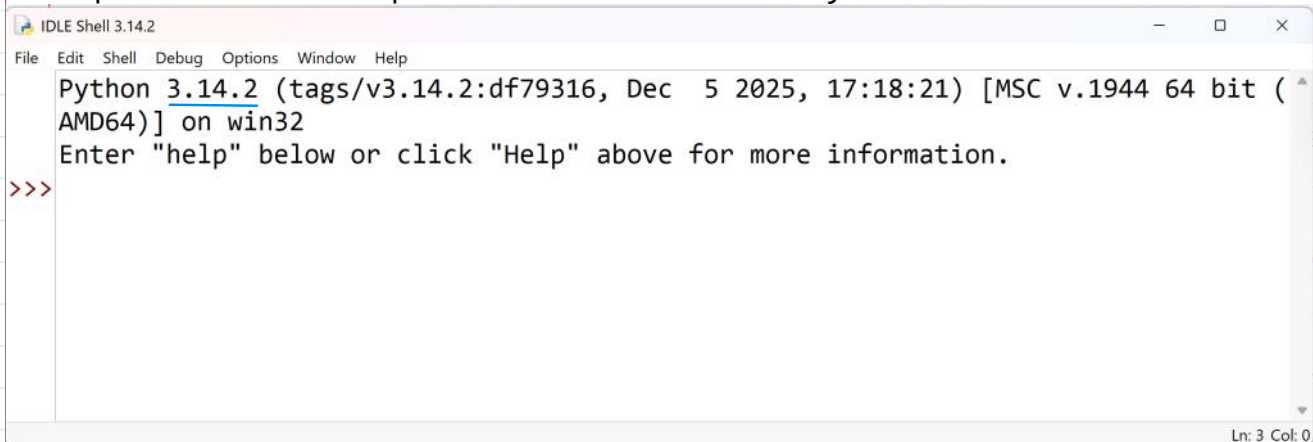
An ordered set of instructions to be executed by computer to carry out a specific task is called a program, and the language used to specify this set of instructions to the computer is called a programming language.

Python is a programming language.

- Python is a high level language. It is a free and open source language.
- It is an interpreted language, as Python programs are executed by an interpreter.
- Python programs are easy to understand as they have clearly defined syntax and relatively simple structure.
- Python is case sensitive.
- Python is portable and platform independent means it can run on various operating systems and hardware platforms.
- Python has a rich library of pre-defined functions.
- Python uses indentation for blocks and nested blocks.

Working with Python

To ride and run (execute) of Python, program, we need to have a Python interpreter installed in our computer or we can use any online Python. Interpreter. The interpreter is also called a Python shell.



Python shell

Execution Modes

There are two ways to use Python Interpreter:

1. Interactive mode
2. Script mode

Interactive mode allows execution of individual statement instantaneously. Whereas script mode allows us to write more than one instructions in a file called Python source code file that can be executed.

1. Interactive Mode: To work in an interactive mode, we can simply type the Python statement on the prompt directly. As soon as we press enter, the interpreter executes the statements and displays the results. As shown below.

```
IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> print('Hello! Python')
Hello! Python
>>> 3456*23+90
79578
>>>
```

Working in the interactive mode is convenient for testing a single line code for instant execution. But in interactive mode, we cannot save the statements for future use and we have to retype the statements to run them again.

Script mode: In the script mode, we can write a Python program in a file, save it and then use the interpreter to execute it. Python scripts are saved as files where filename has extension `.py` By default. Scripts are saved in the Python installation folder. To execute a script we can either:

- i. Type the file name along with the path at the prompt.
- ii. While working in the script mode after saving the file. Or we can click on the play button to execute the code.

Python Keywords

Keywords are reserved words. Each keyword has a specific meaning to the Python interpreter, and we can use a keyword in our program only for the purpose for which it has been defined. As Python is case sensitive. Keywords must be written exactly as it is.

key_list

```
['False', 'None', 'True', 'and', 'as', 'assert', 'async', 'await', 'break', 'class', 'continue', 'def', 'del', 'elif', 'else', 'except', 'finally', 'for', 'from', 'global', 'if', 'import', 'in', 'is', 'lambda', 'nonlocal', 'not', 'or', 'pass', 'raise', 'return', 'try', 'while', 'with', 'yield']
```

```
import keyword
```

```
key_list = keyword.kwlist
```

```
len(key_list)
```

```
35 ← Key values
```

Identifiers

In programming languages, identifiers are names used to identify a variable, function, or other entities in a program. The rules for naming an identifier in Python are as follows:

- The name should begin with an uppercase or a lowercase alphabet, or an `_` sign. This may be followed by any combination of characters `a-z`, `A-Z`, `0-9` or underscore(`_`). Thus an identifier cannot start with a digit.
- It can be of any length. (However, it is preferred to keep it short and

meaningful)

- It should not be a keyword or reserved word.
- We cannot use special symbols like !, @, #, \$, %, ETC. in identifiers.

Example:

```
area = length * breadth
```

Variables: A variable in a program is unity identified by a name (identifier) Variable in Python refers to an object.- An item or element that is stored in the memory. Value of a variable can be string, int, float, complex, bool, or any other object.

Python is a dynamically typed language. In Python we can use an assignment statement to create a new variables and assign specific values to them.

For example:

```
x = 10
type(x)
<class 'int'>
x = 'Aarav'
x
'Aarav'
type(x)
<class 'str'>
x = 5.7
x
5.7
type(x)
<class 'float'>
```

Comments

Comments are used to add a remark or a note. In the source code. Comments are not executed by the interpreter. They are added with the purpose of making the source code easier for humans to understand. They are used primarily to document the meaning and purpose of source code. And its input and output requirement so that we can remember later how it functions and how to use it. For large and complex software, it may require programmers to work in teams, and sometimes a program written by one programmer is required to used or maintained by another programmer. In such situation, documentation in the form of comments are needed to understand the working of the program.

In Python, a comment starts with hash sign(#). Everything follows the hash sign till the end of that line is treated as a comment and the interpreter simply ignores it while executing the statement.

Example:

```
# Variable amount is total spending on grocery
amount = 1000

test1 = 10
test2= 7
test3 = 8
```

```

finalTest = 18
# Total marks is the sum of marks in all the tests of mathematics.
total_marks = test1 + test2 + test3 + finalTest

```

Everything is an Object

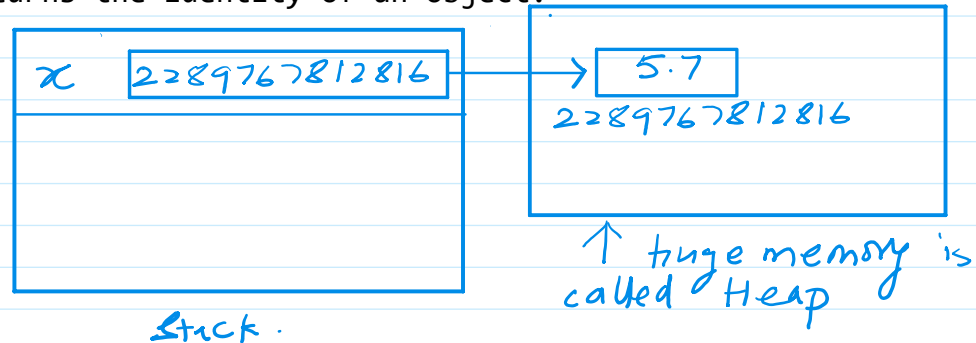
Python treats every value or data item with a numeric, string or other type as an object in the sense that it can be assigned to some variable or can be passed the function as an argument.

Every object in Python is assigned a unique identity (ID) which remains same for the lifetime Of that object. This ID is akin to the memory address of the object. The function ID () returns the identity of an object.

```

x
5.7
id(x)
2289767812816

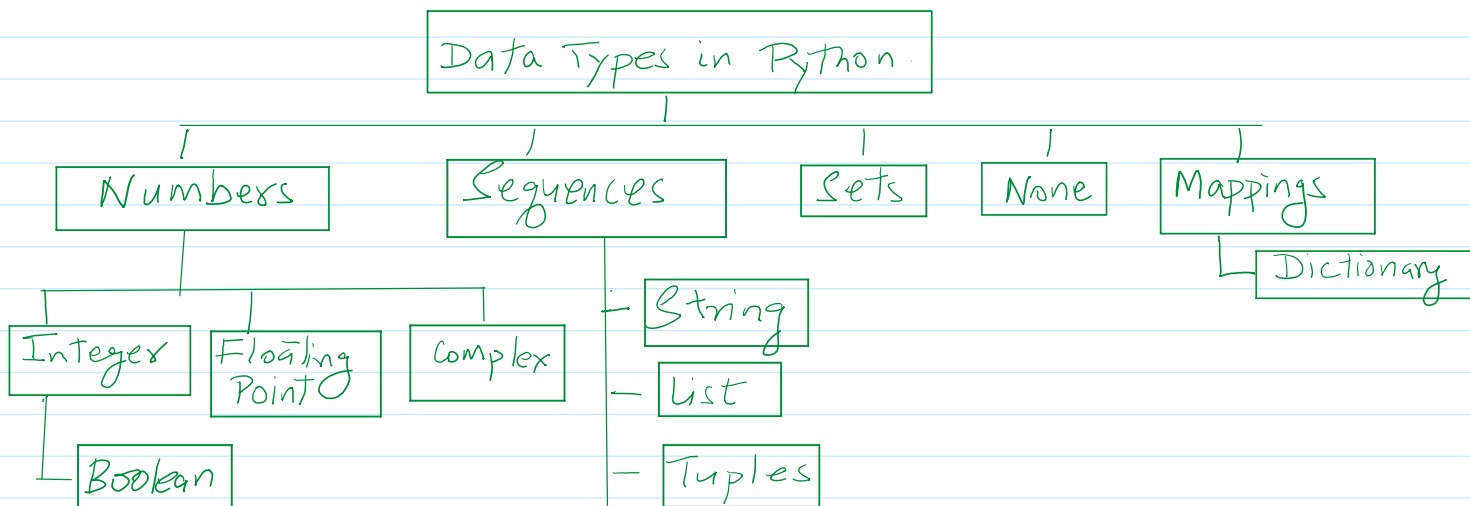
```



Python also comes under the category of Object Oriented programming. However, in Python the definition of object is loosely casted as some objects may not have attributes or other may not have methods.

Data types

Every value belongs to a specific data type in Python. Data types defined the type of data. Values in a variable can hold the operations that can be performed on that data.



Different data type in Python.

History of Python

Guido van Rossum is the creator and founder of the Python programming language.

Guido van Rossum, a Dutch programmer born on January 31, 1956, developed Python in the late 1980s while working at the Centrum Wiskunde & Informatica (CWI) in the Netherlands as a successor to the ABC programming language

W

[Wikipedia+1](#). Python was officially first released in 1991, designed with a focus on code readability and simplicity, allowing developers to express concepts in fewer lines compared to languages like C++ or Java



[GeeksforGeeks+1](#).

Van Rossum served as Python's "**benevolent dictator for life**" (BDFL), overseeing the language's development and making key decisions until he stepped down on July 12, 2018

[Wikipedia+1](#). Afterward, the Python Steering Council took over leadership responsibilities. The language's name was inspired by the British comedy group **Monty Python**, reflecting van Rossum's sense of humor and influence on Python's culture



[python3.info](#).

Throughout his career, van Rossum contributed to other projects, including the ABC programming language, BSD Unix routines, and early web browsers like Grail, and has worked at organizations such as Google, Dropbox, and Microsoft

[Wikipedia+2](#). Python has since become a widely used, high-level, general-purpose programming language, employed in fields ranging from web development and scientific computing to machine learning and education